← Back

Knowledge Check: Deeper Neural Networks
Practice Assignment • 10 min

Due Oct 26, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Why are deep learning models often *non-interpretable*? 1 / 1 point

- (•) They involve complex transformations that are hard to trace
- ( ) They are trained only on random noise
- ( ) They only work with audio data
- ( ) They discard their learned parameters after training

✓ **Nice work**

Correct! Deep learning models perform complex transformations, making them difficult to interpret.

2. What is a key characteristic of a *Recurrent Neural Network (RNN)*? 1 / 1 point

- ( ) It uses dropout instead of activation functions
- ( ) It performs image classification only
- (•) It hasFeedback links and memory of previous states
- ( ) It uses attention layers exclusively

✓ **Nice work**

Correct! RNNs haveFeedback links and memory, allowing them to model sequences.

3. What is the role of *hidden layers* in a deep neural network? 1 / 1 point

- ( ) They act as the input features
- (•) They transform and learn feature representations between input and output
- ( ) They delete irrelevant input tokens
- ( ) They store final output labels

✓ **Nice work**

Correct! Hidden layers learn intermediate representations between input and output.

4. What is the benefit of using a *bidirectional LSTM* over a regular LSTM? 1 / 1 point

- ( ) It eliminates activation functions
- (•) It processes the sequence both forward and backward
- ( ) It increases the number of neurons per layer
- ( ) It removes the need for training data

✓ **Nice work**

Correct! A bidirectional LSTM processes the sequence in both directions.

5. Which neural network architecture uses *convolution* instead of matrix multiplication? 1 / 1 point

- (•) Convolutional Neural Network (CNN)
- ( ) Multi-Layer Perceptron (MLP)
- ( ) Conditional Random Field (CRF)
- ( ) Recurrent Neural Network (RNN)

✓ **Nice work**

Correct! CNNs use convolution (dot product) operations instead of matrix multiplication.